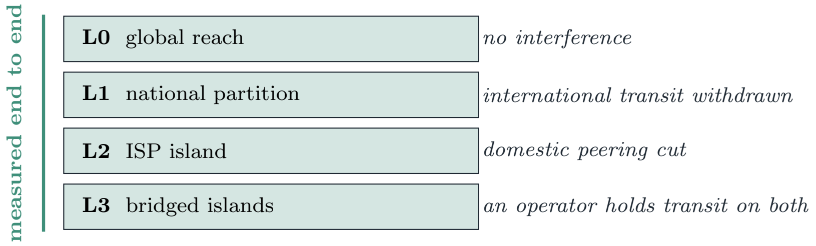
Diagram workbench

Figures for the NSysS submission, drawn full size. Nothing here is page-limited.

Figure A — Resilience ladder, reframed to L0–L3

Replaces the L0–L5 ladder in main.tex. The mesh and radio rungs are gone: this system is a forum that survives blocking and blackout over IP, and off-grid transport is explicitly not claimed. The right column names who cuts each rung, which is what makes the ladder a threat statement and not a feature list.



Out of scope: off-grid mesh and long-range radio. The system needs an IP path; it does not need an unfiltered *one.*

Figure B — One write path, and where an envelope goes

The architectural claim in one picture: a single write endpoint, one pipeline, and three consumers of the same accepted envelope (projection, witness log, outbox). Clients never speak the federation protocol.

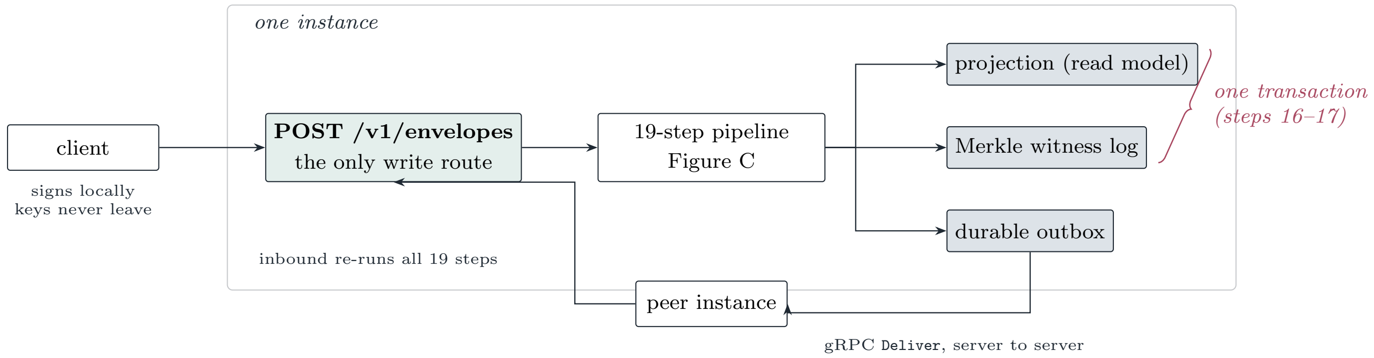
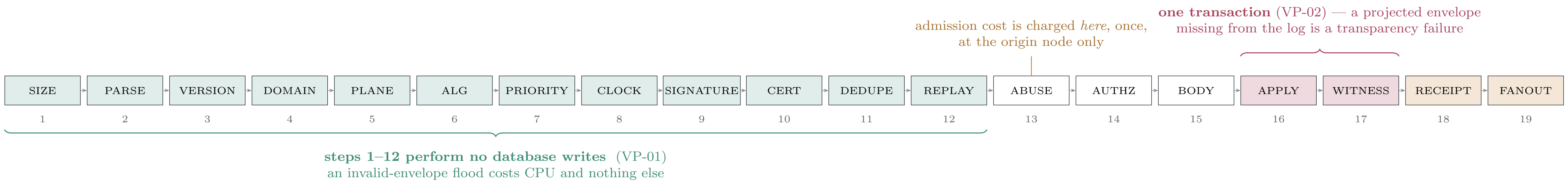


Figure C — The nineteen-step validation pipeline

The paper names this pipeline a dozen times and never shows it. Two orderings carry the argument and both are visible here: nothing before step 13 touches the database (so an invalid flood cannot amplify into writes), and 16–17 commit together (so nothing can be visible in a projection while missing from the log).



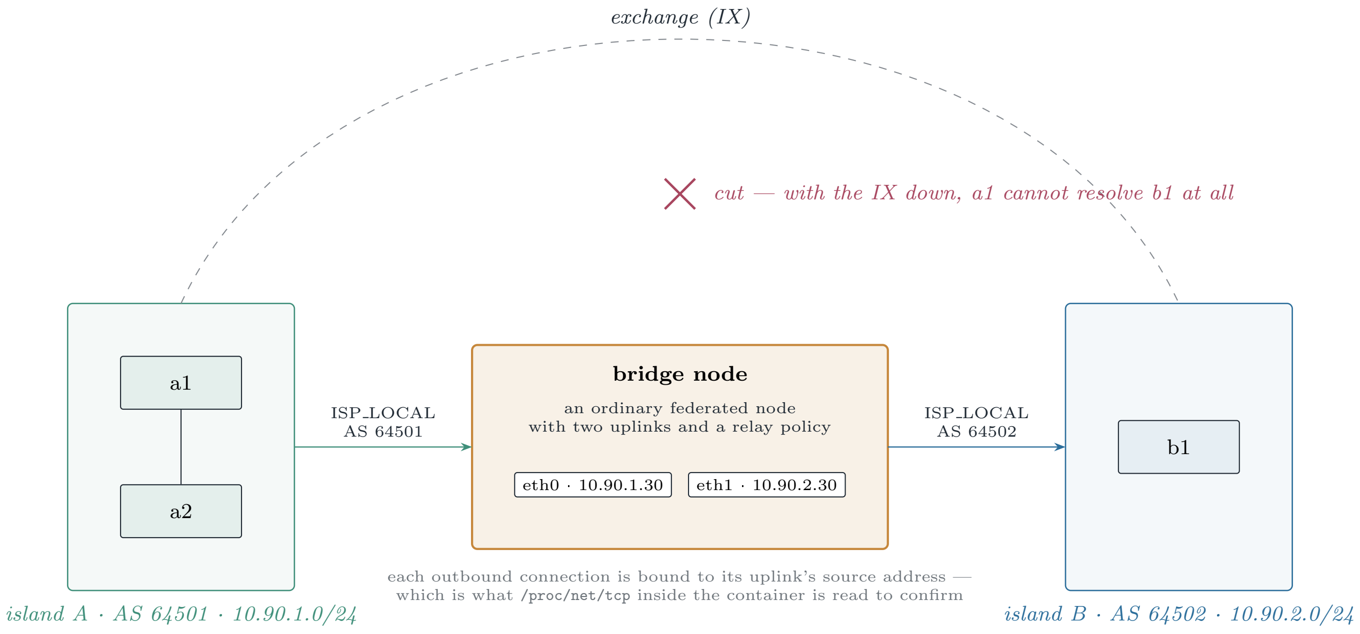
1 SIZE — on raw bytes, before parsing
2 PARSE — canonical decode, unknown fields rejected
3 VERSION — known envelope version
4 DOMAIN — must exist in the registry
5 PLANE — registry plane equals envelope plane
6 ALG POLICY — key algorithm permitted here
7 PRIORITY — equals the declared class
8 CLOCK — inside [now−max_age, now+max_skew]
9 SIGNATURE — Ed25519 over canonical fields 1–12
10 CERTIFICATE — valid and unrevoked at created_at_ms
11 DEDUPE — unique index on (content_id, direction)
12 REPLAY — (author_key, nonce) unseen
13 ANTI-ABUSE — credential / nullifier / PoW / credits
14 AUTHORISE — permission checked against the projection
15 BODY VALIDATE — domain-specific field constraints
16 APPLY — project into the read model
17 WITNESS — append the content id to the Merkle log
18 RECEIPT — sign and return
19 FANOUT — enqueue by class, never back to the origin

A federated envelope does not get a shorter pipeline. All nineteen steps re-run at the receiver, from every transport, so a trusted peer's forged envelope is refused exactly as a stranger's is. Step 13 is the single exception, and for a precise reason: proof of work, credits, blind credentials and epoch nullifiers are each keyed to *one* node, so a proof minted at the origin is not merely unverifiable elsewhere, it is meaningless elsewhere. Charging again on receipt would make every gated domain impossible to federate. What is skipped is a *payment*, not a check; the receiver's protection is the per-peer, per-class quota instead.

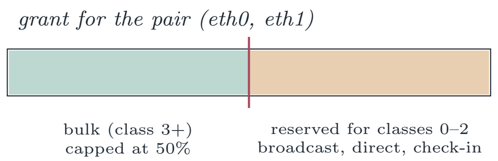
Why the ordering is the security property. Steps 1–12 answer every question that can be answered from the bytes and the local index alone. Only after all of them have passed does the node touch its database. Measured against 3000 well-formed envelopes carrying a mutated byte inside the signed region — novel content ids, so deduplication cannot short-circuit them — the node absorbed 730 req/s and its database recorded **zero writes**. The honest qualification is that 90% of that flood was shed by the per-peer rate limiter before reaching signature verification, so the no-write prefix is the *second* line of defence, not the first.

Figure D — ISP bridging: topology, source binding and quota

*The L3 mechanism at full detail. Two islands on separate ASNs, the exchange between them cut, and one node holding transit on both. The bridge binds each outbound federation connection to the source address of the matching uplink, which is what **`/proc/net/tcp`** is read to confirm.*



Accounting is per uplink *pair*, not per peer (BR-04). An island with three nodes is charged once for one link's traffic, not three times — charging per peer would make an island's cost depend on how many volunteers happened to stand up nodes there. Within a pair, classes 0–2 hold reserved capacity and bulk draws on a bucket capped at half the pair grant, so a forum backlog can never starve an emergency broadcast.



Bridging rules, as implemented

- BR-01 Opt-in per node, and requires TRUSTED status with at least one peer on *each* side. An untrusted node cannot volunteer itself as the chokepoint between two islands.
- BR-02 A relayed envelope runs the **full** nineteen-step pipeline before relay. The bridge is a verifying relay, not a repeater.
- BR-03 Never relay back to the uplink it arrived on. Content dedupe would terminate the loop anyway, but not before it had cost a round trip.
- BR-04 Relay quotas are per uplink pair and per class, with classes 0–2 reserved.
- BR-05 The bridge advertises `is_bridge` and `bridged_asns` in its peer record, so peers know a cross-island path exists and can prefer it during a partition.
- BR-06 Bytes relayed per direction, per class and per hour, plus current quota headroom, are observable to the operator.
- BR-07 An uplink state change re-evaluates *every* active peer path within 30 s; streams on a dead uplink are torn down and re-established on a live one.
- BR-08 Switching uplinks loses nothing queued. The outbound queue is durable and uplink-agnostic, so only the path changes.
- BR-09 After a switch, affected peers are re-`Announce`d with the new endpoint set and `Backfill` closes any gap the transition opened.
- BR-10 An operator override can force an uplink up or down, or pin a peer to a specific uplink, for the conditions probes cannot detect.

Figure E — Sequence: publish once, federate to a peer

The ordinary path, with the two things reviewers ask about marked: the receipt is returned before fanout, and the peer re-runs every validation step.

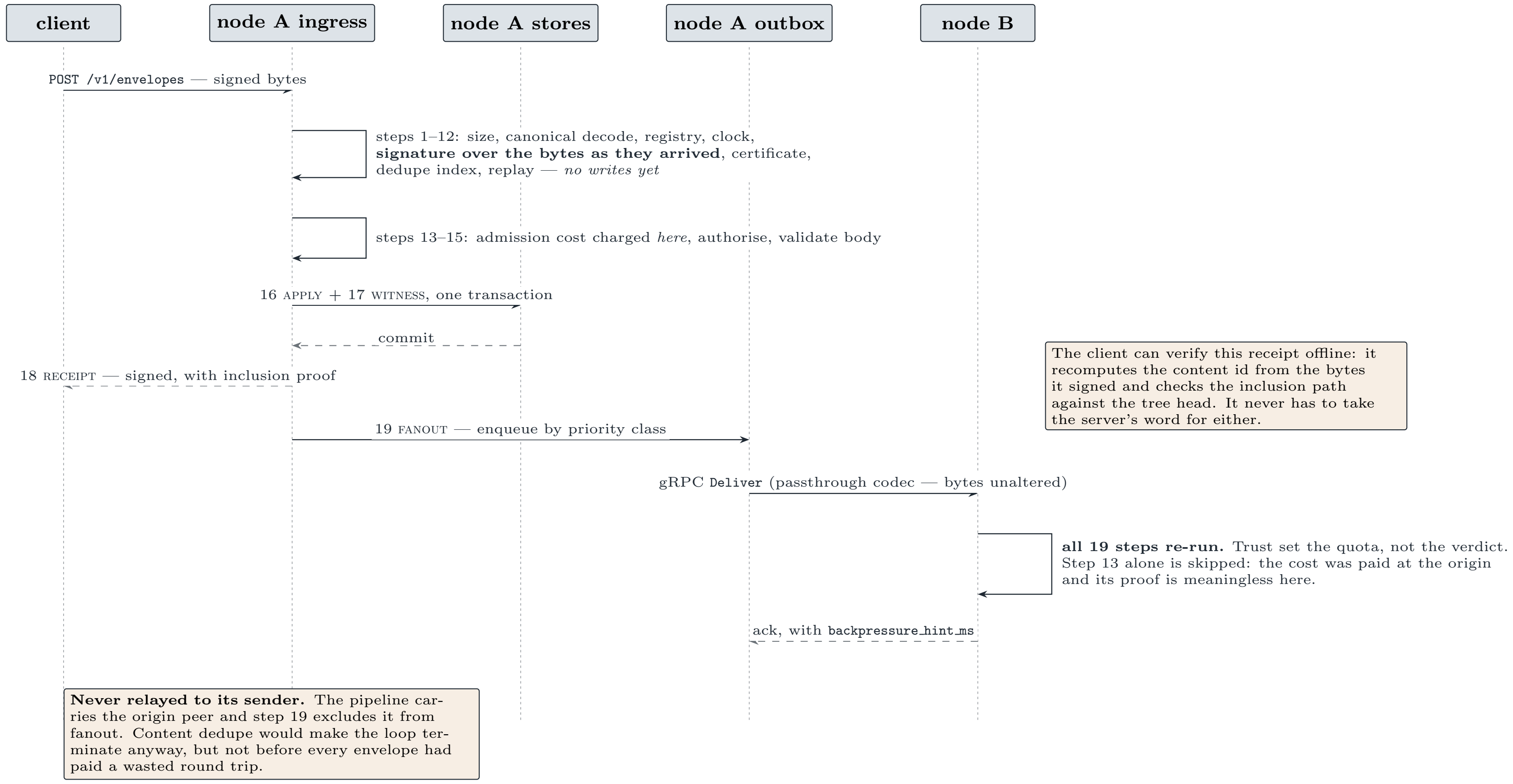


Figure F — Sequence: a crossing after the exchange is cut

Why a "crossing" is three envelopes and not one. A post that island B can actually render needs the author's certificate and the community record first, so the measured cost is the whole dependency chain — 0.5 s, 1.2 s, 2.1 s.

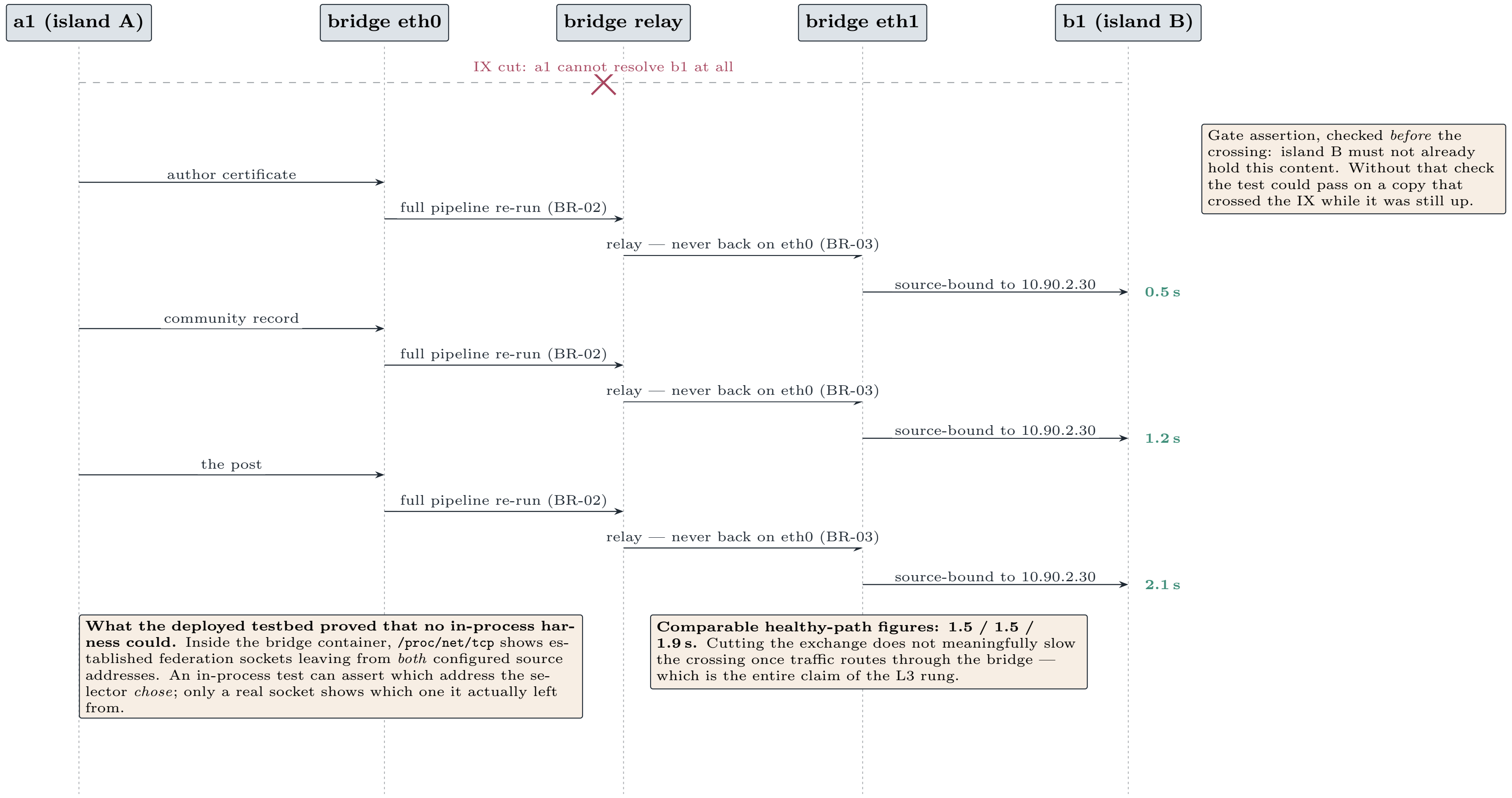


Figure G — Sequence: uplink failover without loss

BR-07 through BR-10 as one timeline. The property worth drawing is that the queue is uplink-agnostic, so a switch changes the path and nothing else.

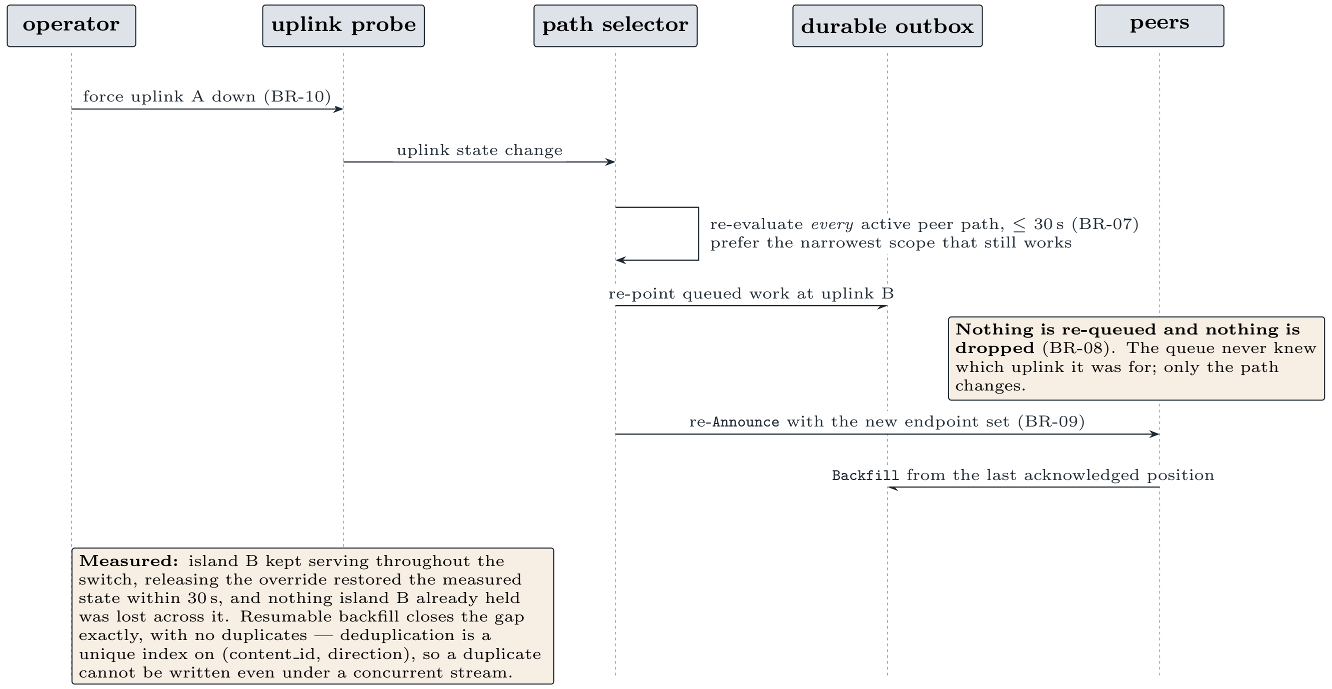


Figure H — Sequence: a stale tree head is not evidence of a fork

The defect and the fix in one diagram. Four code paths sample a peer's head and nothing orders their arrival, so an older reading routinely lands after a fresher one. Compared pairwise with no consistency proof, that reads as a log rollback.

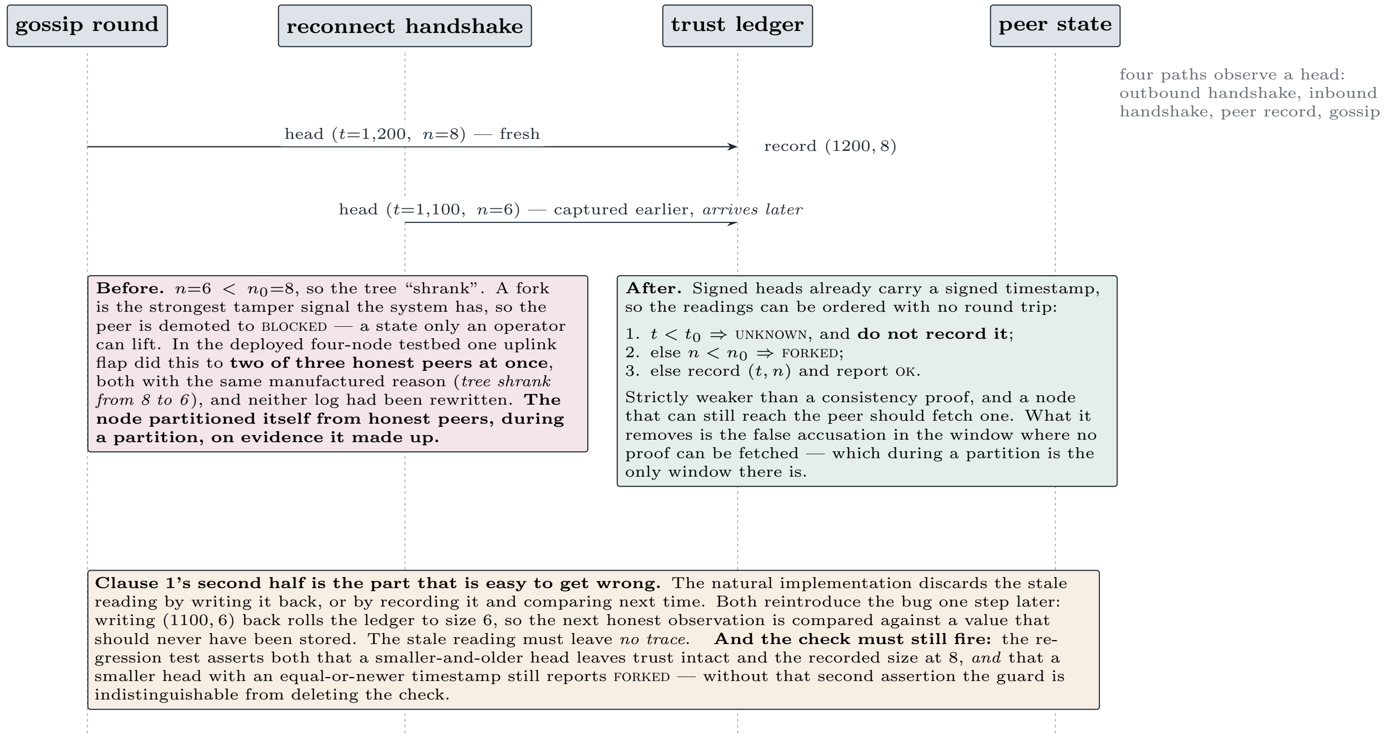
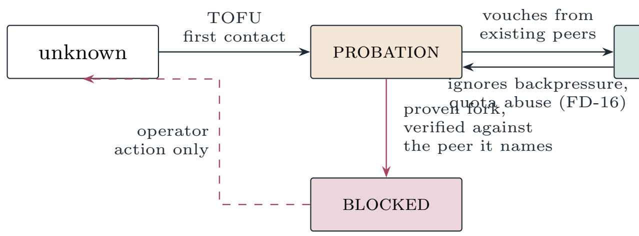


Figure I — Peer admission: TOFU, vouching, and the one state an operator must lift

Why there is no admin allowlist: during a shutdown, volunteers stand up relay nodes and cannot wait for manual approval. New peers are admitted immediately at a low quota and earn reach.



Two rules keep this from becoming a censorshi itself.

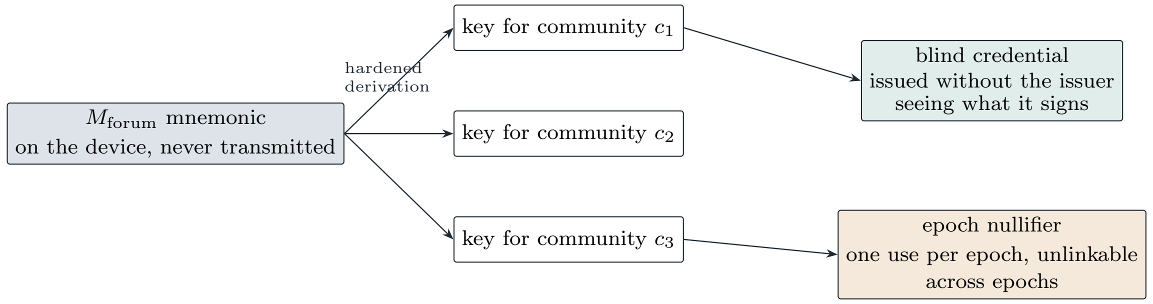
1. A claim *about* a peer arriving *from* another peer i dence. Tree-head observations are relayed labelled b layer, so any check that can BLOCK must first verify t belongs to the peer it names — otherwise one peer c lence another.

2. Bridging requires TRUSTED on *both* sides (BR-01). An untrusted node cannot volunteer itself as the chokepoint between two islands.

What is published, and what is not. The peer directory and `ExchangeDirectory` *omit* blocked peers rather than labelling them: a directory naming who a node blocked is a target list for whoever wanted them blocked. The *finding* is published instead — the alerts endpoint names the peer and the reason — because evidence of censorship is only useful if someone can see it.

Figure J — Pseudonymity without a server: key derivation and admission cost

What makes the forum anonymous rather than merely federated. No server ever learns which pseudonyms belong to one person, and spam is priced rather than policed by identity.



The property this buys. A moderator can act on a pseudonym, a peer can rate-limit it, and a credential can prove "this author is entitled to post here" — while no party, including the node that issued the credential, can link two of one person's community identities or link any of them to a device or a phone number.

Why identity is not the anti-abuse primitive. Requiring a verified identity to stop spam hands the adversary exactly the lever it wants: a list of who is speaking, and a chokepoint at which to refuse them. Cost is used instead — memory-hard proof of work, credits, blind credentials, epoch nullifiers — all of which work against a fully anonymous participant. The honest limitation is that cost bounds the *rate* of abuse, not the *number* of identities, and we state that rather than claiming Sybil resistance we do not have.

And a check-in costs nothing. Telling people you are alive is not rate-limited and needs no credential. A design that charges for that has misunderstood which message matters most during a shutdown.

Figure K — Threat model on one axis: what is resisted, and what is not

A compact version of §5 for reviewers who read figures first. The left column is gated in the implementation; the right column is deliberately unbought.

Resisted, and where

Attack	**The property that resists it**
A server suppresses content by withholding approval	Publish-then-attest: content is valid the instant its author signs it, so approval is removed as a mechanism rather than policed. Moderation is an additive signed opinion a client may honour.
A relay tampers with content in flight	The signature covers *the bytes as they arrived*; no intermediary re-encodes them; a non-canonical encoding is rejected, never repaired.
A peer rewrites its own history	Signed tree heads gossiped between peers, with fork detection that stays safe when no consistency proof can be fetched.
An invalid-envelope flood amplifies into database load	Steps 1–12 perform no writes. Measured: 730 req/s absorbed, zero writes.
An ISP- or nation-level partition cuts reach	Continuous preference for the narrowest working scope, plus a multi-homed bridge between islands.
One peer frames another to get it silenced	A check that can BLOCK must verify the claim belongs to the peer it names *and* is current. An older reading of remote state is not evidence.

Not resisted, deliberately

No traffic-analysis resistance. No cover traffic, no mixing, no timing defence. An observer on both islands can correlate flows.

No transport metadata protection. An ISP sees that a node federates, roughly with whom, and roughly how much.

No endpoint-compromise resistance. A seized device loses its keys. Revocation exists but only helps once it reaches peers, which is what the adversary is attacking.

No compromised-majority peer set. Fork detection assumes at least one honest observer gossips a conflicting head.

No formal unlinkability proof. The separation between identity planes is architectural, with no anonymity-set analysis.

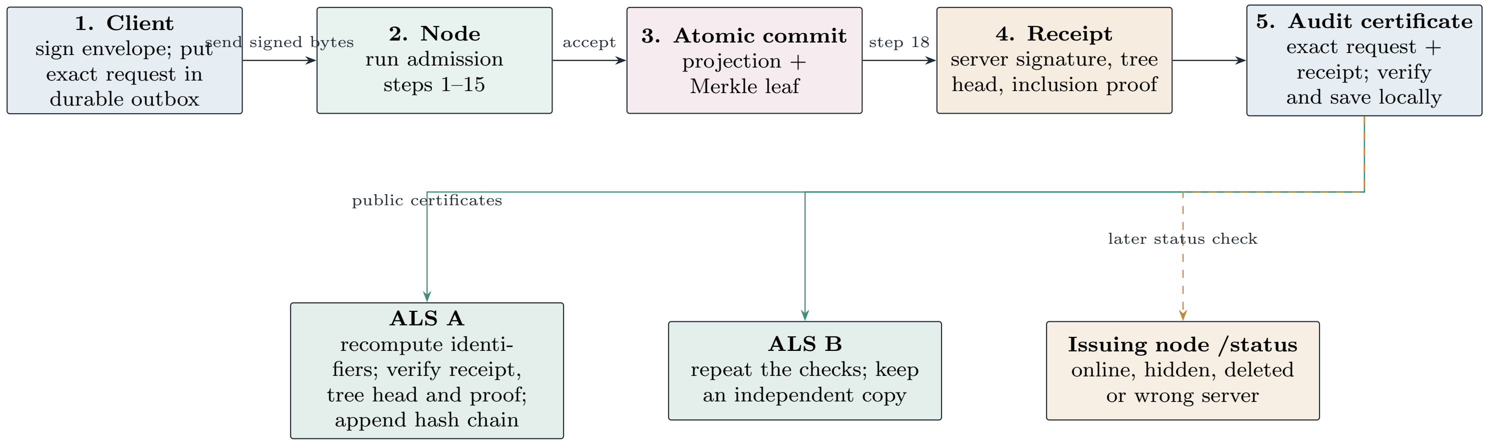
No identity-derived Sybil resistance. Cost bounds the rate of abuse, not the number of identities. This is a choice, and it has a price.

No delivery with every link cut. With no path at all the system stores and forwards; it does not deliver.

Each of these could be bought, and each would be bought with availability — the property the system exists to preserve — or with the ability to run on a single-board computer over an intermittent link. Where a defence and reach conflict, reach wins, and the limit is stated rather than hidden.

Figure L — Audit-log server workflow: acknowledgement becomes portable evidence

The audit-log server does not moderate content or report the node's live state. It independently checks and preserves proof that a node acknowledged a particular signed request.



Store first, forward second. If audit delivery fails during a blackout, the locally saved certificate remains pending and can be forwarded when any route returns. Exact retries are idempotent. A conflicting certificate for the same server and content pair is rejected.

The boundary matters. The ALS proves only that the issuing node acknowledged the request at a particular logged position. It cannot prove acceptance without a receipt, compel the node to keep serving the body, or know whether that body is currently hidden or deleted. The issuing node answers that last question. Direct-message certificates stay on the client because exporting them would create a timestamped social graph.